

CLAIMS

Claim 1:

A computer-implemented system for adaptive network configuration deployment, comprising:

a physical metric collector configured to collect device physical state metrics;

a damping factor calculation module configured to calculate a damping factor based on the collected physical state metrics;

a deployment scheduler configured to dynamically adjust deployment concurrency and execution timing based on the damping factor;

and a deployment execution module configured to perform configuration deployment across network devices.

Claim 2:

The system of Claim 1, wherein the damping factor increases when device temperature, CPU load, or hardware aging increases.

Claim 3:

The system of Claim 1, wherein deployment concurrency is inversely proportional to the damping factor.

Claim 4:

The system of Claim 1, wherein deployment delay between execution steps is proportional to the damping factor.

Claim 5:

The system of Claim 1, wherein the system determines convergence based on a residual representing the difference between actual device state and target configuration state.

Claim 6:

A computer-implemented method for adaptive network deployment, comprising:

collecting device physical state metrics;

calculating a damping factor based on the collected metrics;

adjusting deployment concurrency based on the damping factor;

deploying configuration to network devices;

monitoring convergence using a residual;

and completing deployment when the residual falls below a convergence threshold.